



CMPT 413/713: Natural Language Processing

Project information

Spring 2025
2025-02-12

Summary of topics

- So far
 - Language Modeling: $P(w_i | w_{1:i-1})$
 - Text classification: $P(c | d)$
 - with Naive Bayes, Logistic Regression and Neural Networks
 - Static and contextual word embeddings: Representing w as a vector
 - Sequence modeling with RNNs and Transformers
- After the break
 - Focus on project
 - More on pretraining and fine-tuning LLMs
 - Structured representations and parsing
 - Other NLP applications and using LLMs

Project

- Project timeline
 - 2-20 - Initial abstract/title (not graded) for initial feedback
 - 2-24 - Proposal due
 - 3-19 - Project milestone due
 - 4-6 - Final project videos
 - 4-10 - Final project report due

Project Proposal

- Initial title and abstract (~250 words)
 - Think about what you want to do for your project
 - Have an informative title (“CMPT 413 class project” is not a good title)
 - Describe the topic/problem your group will work on
 - Scope of the project
 - Data and compute resources you plan to use
- Proposal (1-2 pages)

<https://angelxuanchang.github.io/nlp-class/project.html>

Project Proposal

- What task are you addressing? What is the input / output? Why is it interesting?
- What specific aspects will your project be on?
 - Re-implement paper? Compare different methods? Analysis?
- What **data** do you plan to use? For training vs for evaluation?
 - Preliminary statistics for your data (number of sentences, tokens, etc)
- What is the specific method or methods you will use to address the task?
 - **What will you implement by yourself vs what existing code will you use?**
 - What **compute resources** do you plan to use?
- How do you plan to evaluate?
 - Data splits?
 - What **metrics**?
 - What experiments will you run to **compare** different variations / different approaches?
- **Timeline and work breakdown**
 - What do you plan to have by the milestone? The end of the term?
 - Who will work on what?

What kind of approaches you want to take

- In this class, we started with fundamentals and simple ML models (Naive Bayes and Logistic regression)
- You should be experimenting with methods beyond that
 - Use pretrained models! Do fine-tuning!
 - Use LLMs with few-shot prompting
 - We will cover more about these topics after the break

Type of problem and resources

<https://angelxuanchang.github.io/nlp-class/project.html>

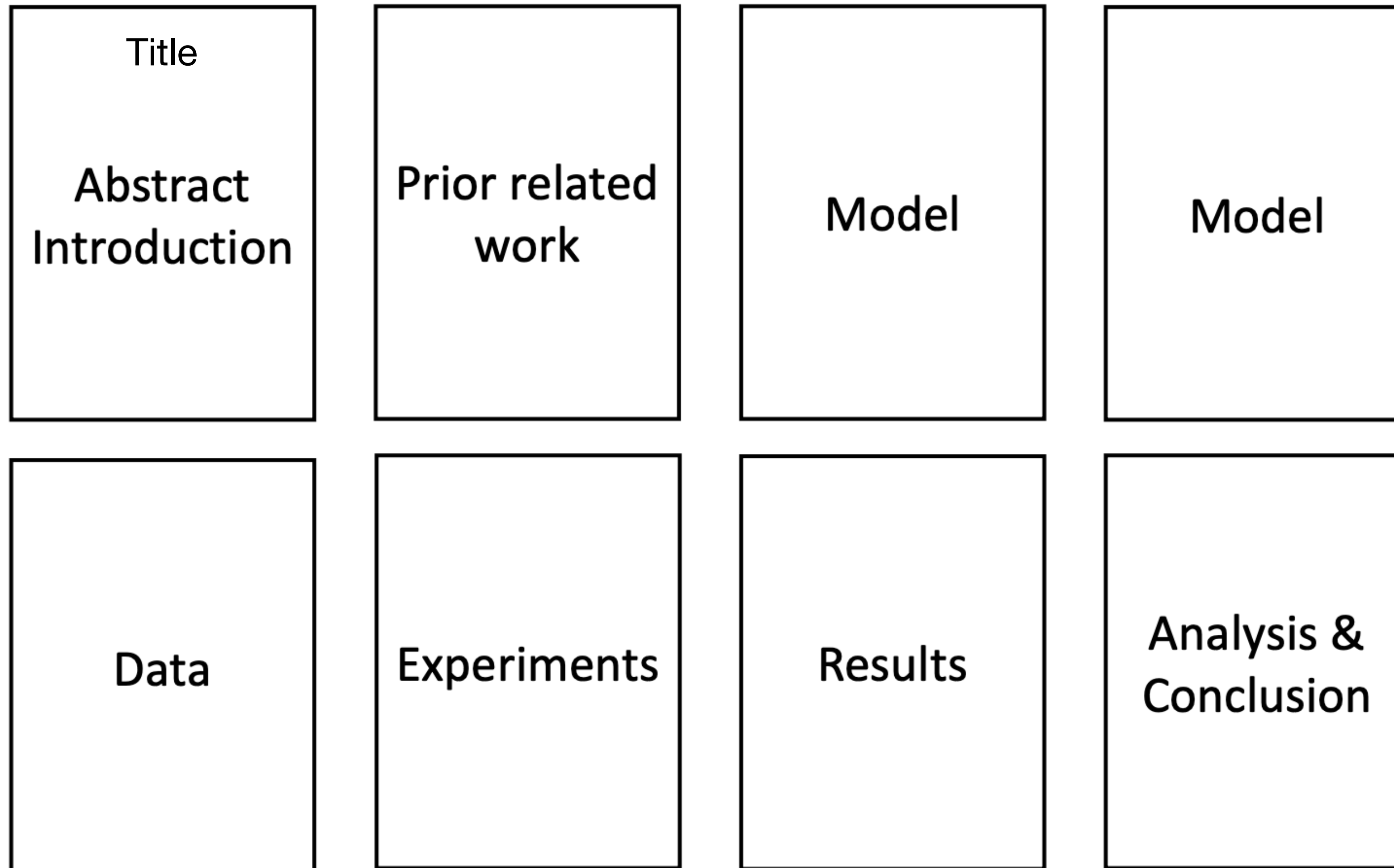
- Pick a problem based on what are you interested in?
- Types of problems
 - Building a cool NLP application
 - Re-implementing / re-reproducing a recent paper
 - Applying an existing neural model to a new task
 - Implementing a complex neural architecture
 - Proposing a new neural model or a new variation of an existing model
 - Proposing a new training, optimization, or evaluation scheme
 - Experimental and/or theoretical analysis of a NLP model
 - Probing the capabilities of current NLP models

Final project report

8 pages not including references

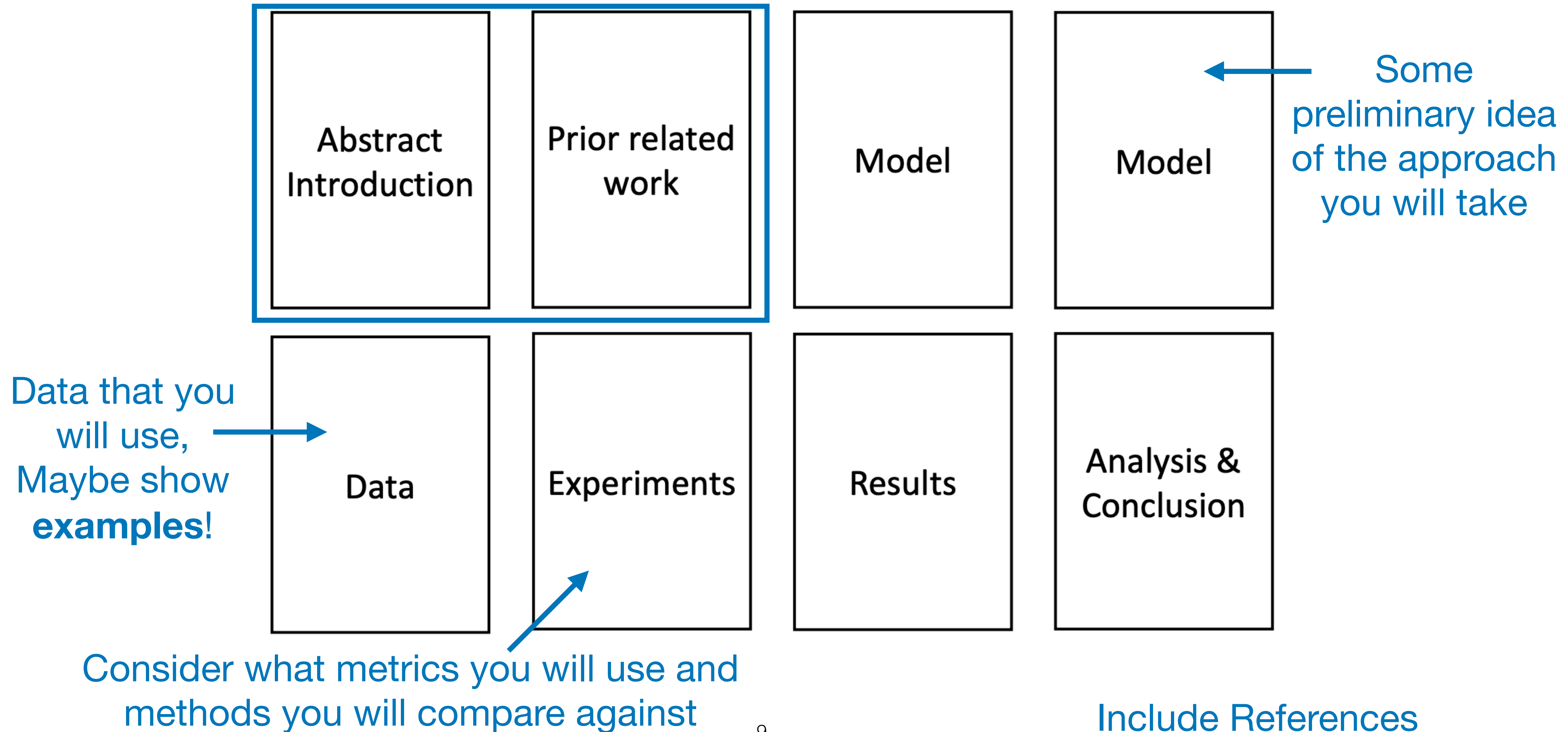
Model or what approach you are taking

Rough page layout:
your report can have
different number of
pages for each
section



What is expected to be included in the project proposal

We expect you to have around 1-2 pages (can be more)



Be careful if you try to use LLMs to help you with your abstract!

Using ChatGPT for abstract writing



You

Please write a one paragraph abstract for a class project that uses BERT for sentiment analysis for toxic comment classification



ChatGPT

Title: BERT-Enhanced Sentiment Analysis for Toxic Comment Classification: A Comprehensive Investigation

Abstract:

This class project delves into the realm of toxic comment classification using Bidirectional Encoder Representations from Transformers (BERT), an advanced natural language processing model. Toxic comments pose significant challenges in online discourse, necessitating robust sentiment analysis tools for content moderation. The project begins by elucidating the theoretical underpinnings of BERT and its capacity for capturing contextualized word embeddings. We focus on the specific application of toxic comment classification, addressing preprocessing steps, dataset selection, and domain-specific challenges associated with online language. Evaluation metrics such as accuracy, precision, recall, and F1 score are employed to assess the effectiveness of BERT in identifying and categorizing toxic comments. Additionally, the interpretability of the model is explored through attention mechanisms, shedding light on key linguistic indicators of toxicity. The project aims to provide insights into the application of BERT for enhancing content moderation efforts, ultimately contributing to a safer and more inclusive online environment.

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Vague and not informative

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Okay

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Additionally, the interpretability of the model is explored through attention mechanisms, shedding light on key linguistic indicators of toxicity. The project aims to provide insights into the application of BERT for enhancing content moderation efforts, ultimately contributing to a safer and more inclusive online environment.

Some NLP Tasks

- SQuAD: question answering
- SNLI: natural language inference
- SRL: semantic role labeling
- Coref: coreference resolution
- NER: named entity recognition
- SST-5: sentiment analysis
- POS: Part-of-speech tagging
- Entity linking
- Constituency Parsing
- Dependency Parsing
- Intent detection and Slot filling
- Machine translation
- Paraphrasing
- Summarization
- Semantic Parsing

Tasks

- SQuAD: question answering
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Stanford CS224 default project

– SQuAD

– RobustQA

Passage

In meteorology, precipitation is any product of the condensation of atmospheric water vapor that falls under **gravity**. The main forms of precipitation include drizzle, rain, sleet, snow, **graupel** and hail... Precipitation forms as smaller droplets coalesce via collision with other rain drops or ice crystals **within a cloud**. Short, intense periods of rain in scattered locations are called “showers”.

Question / Answer

What causes precipitation to fall?

gravity

What is another main form of precipitation besides drizzle, rain, snow, sleet and hail?

graupel

Where do water droplets collide with ice crystals to form precipitation?

within a cloud

SQuAD: 100,000+ Questions for Machine Comprehension of Text

Rajpurkar et al, EMNLP 2016

Tasks

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- SNLI: natural language inference
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Premise

Hypothesis

contradiction

entailment

neutral

A man inspects the uniform of a figure in some East Asian country.

The man is sleeping.

An older and younger man smiling.

Two men are smiling and laughing at the cats playing on the floor.

A soccer game with multiple males playing.

Some men are playing a sport.

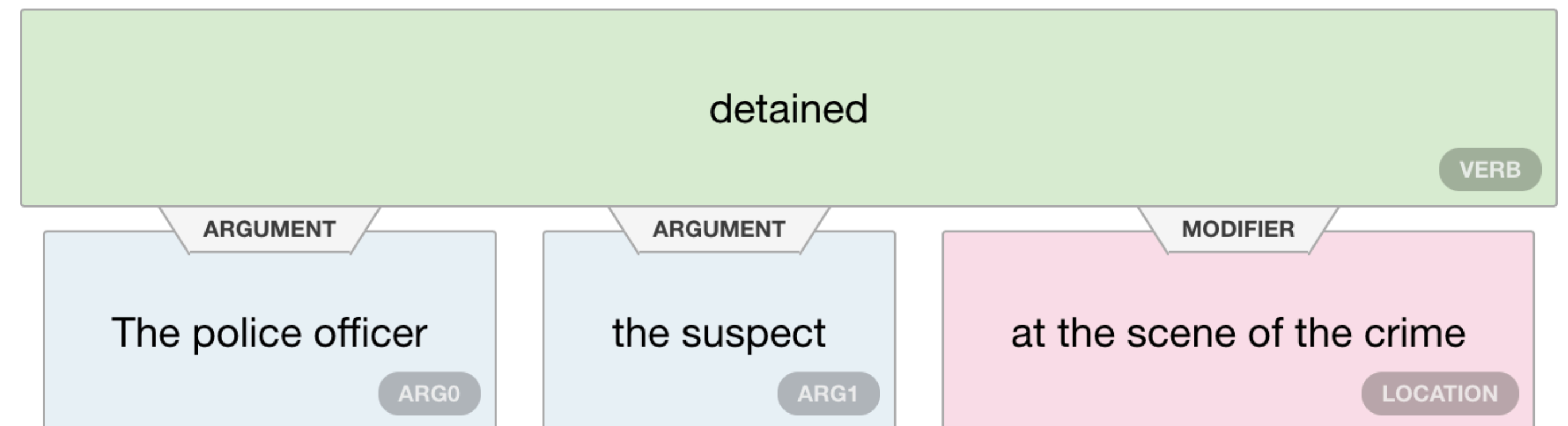
A large annotated corpus for learning natural language inference

Bowman et al, EMNLP 2015

Tasks

- SQuAD: question answering
 - SNLI: natural language inference
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-
- Different types of frames
 - FrameNet vs PropBank
 - Different semantics roles depending on the verb

The police officer detained the suspect at the scene of the crime

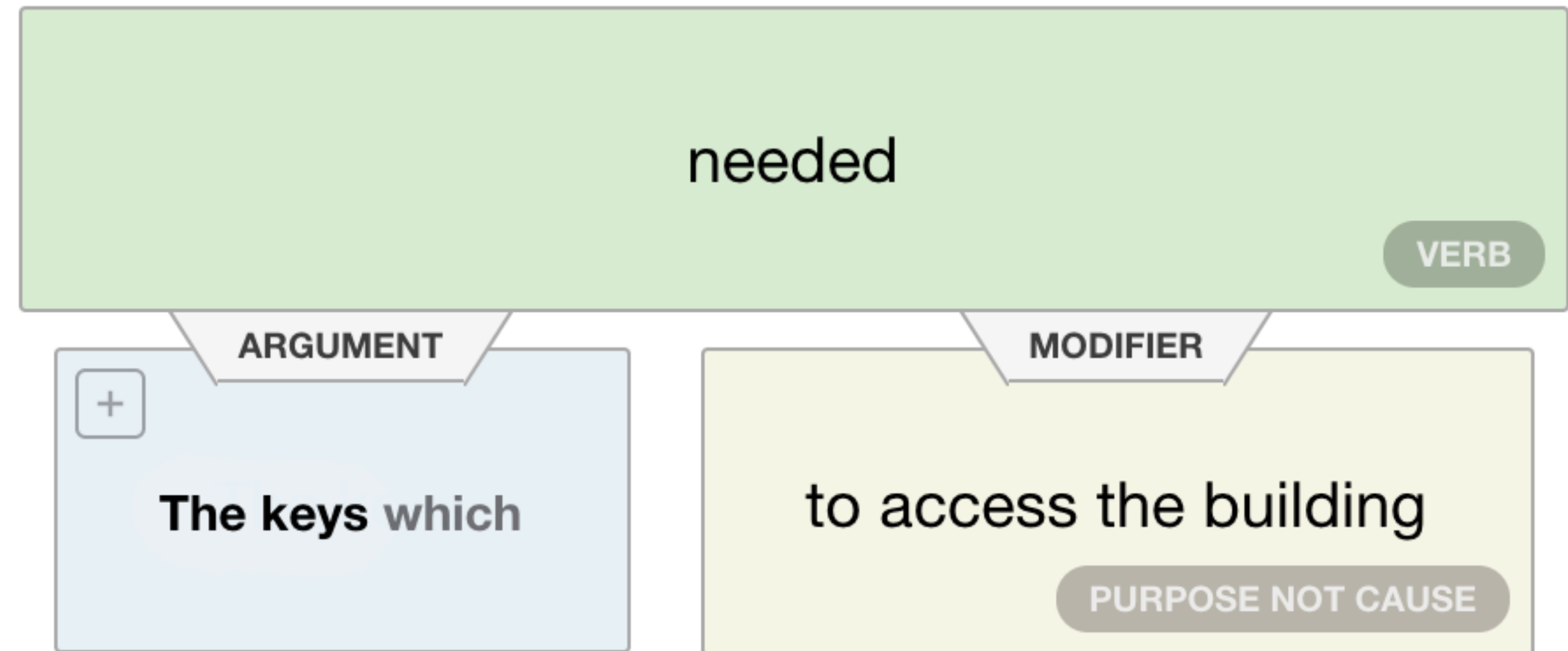


(figure credit: <https://demo.allennlp.org/semantic-role-labeling/>)

Tasks

- SQuAD: question answering
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The **keys**, which were **needed** to access the building, were locked in the car.



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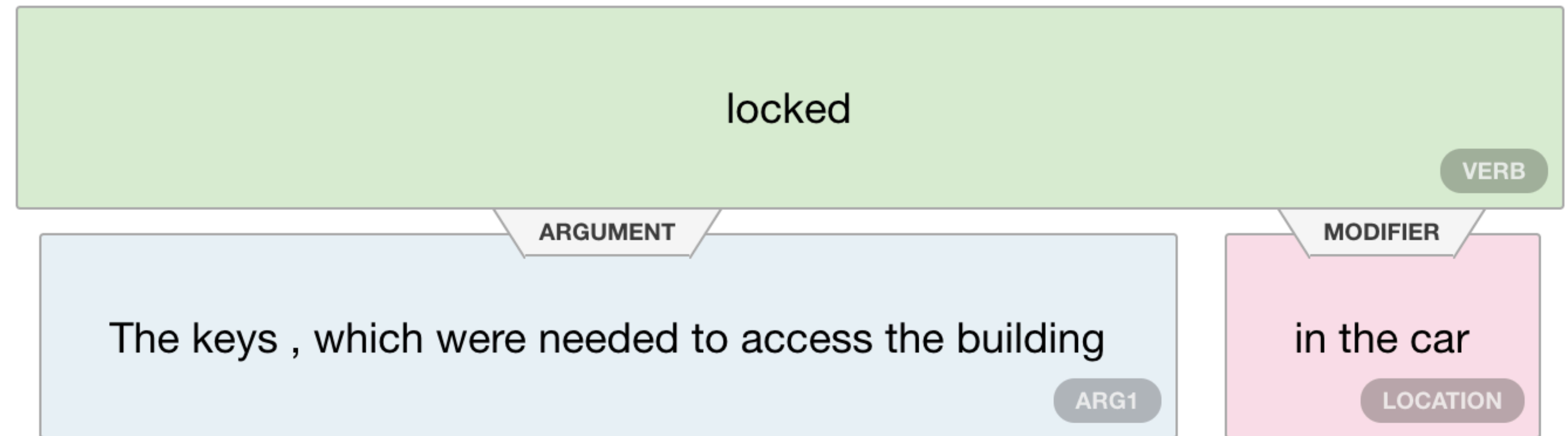


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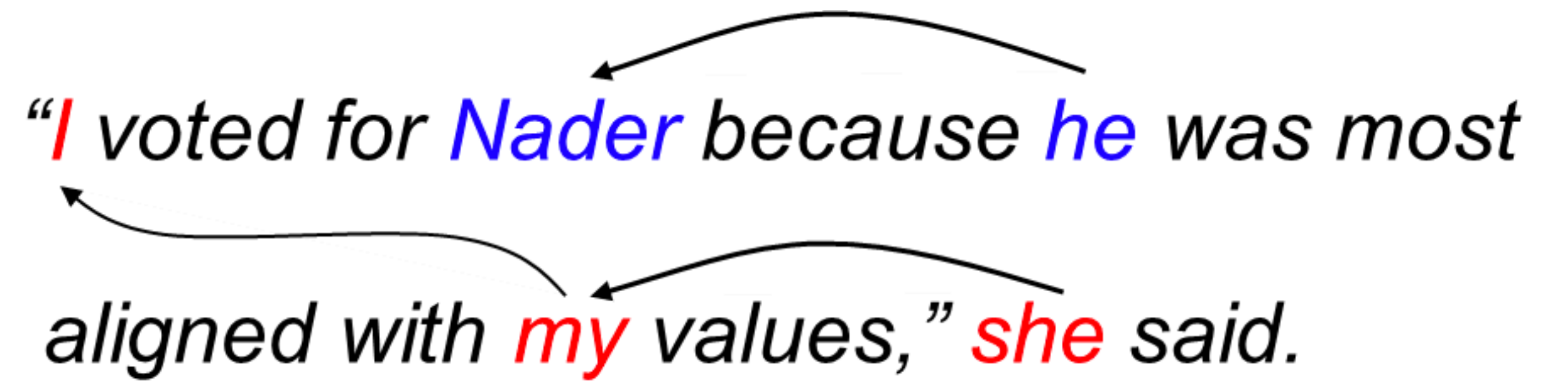
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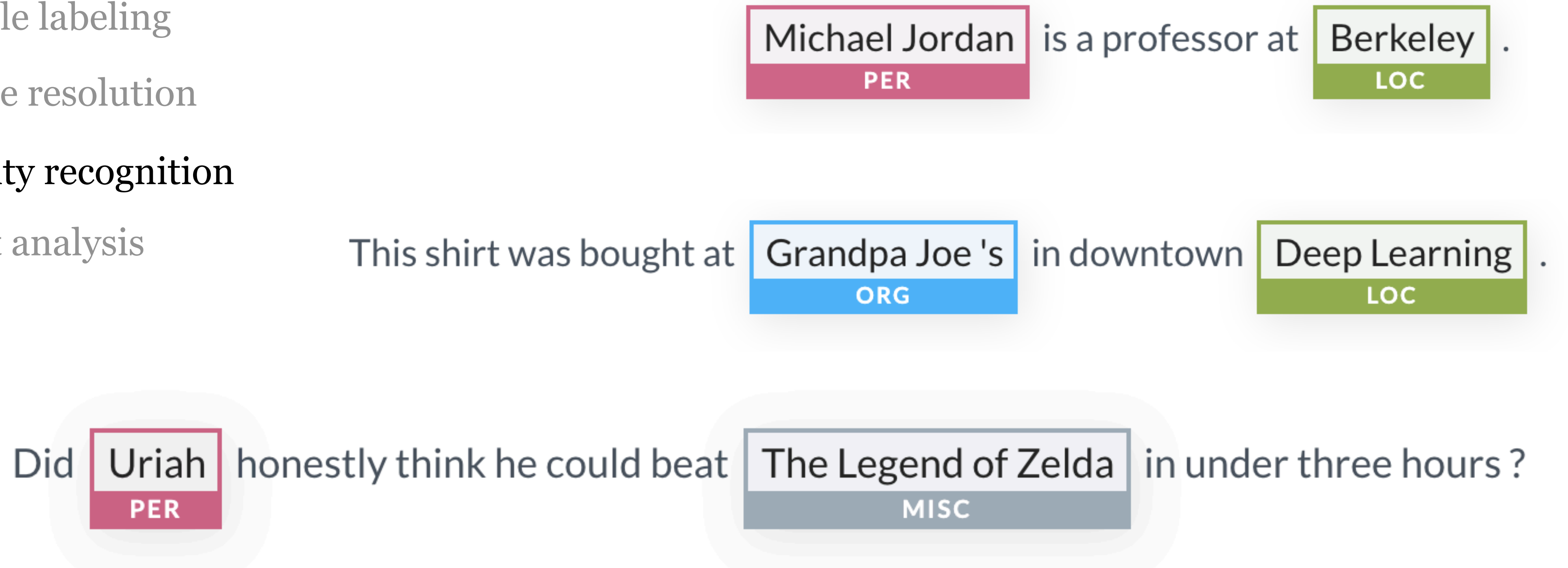


“I voted for Nader because he was most aligned with my values,” she said.

(figure credit: <https://nlp.stanford.edu/projects/coref.shtml>)

Tasks

- SQuAD: question answering
- SNLI: natural language inference
- SRL: semantic role labeling
- Coref: coreference resolution
- **NER: named entity recognition**
- SST-5: sentiment analysis



(figure credit: <https://demo.allennlp.org/named-entity-recognition>)

A large annotated corpus for learning natural language inference

Sang and Meudler, CoNLL Shared Task 2002

Tasks

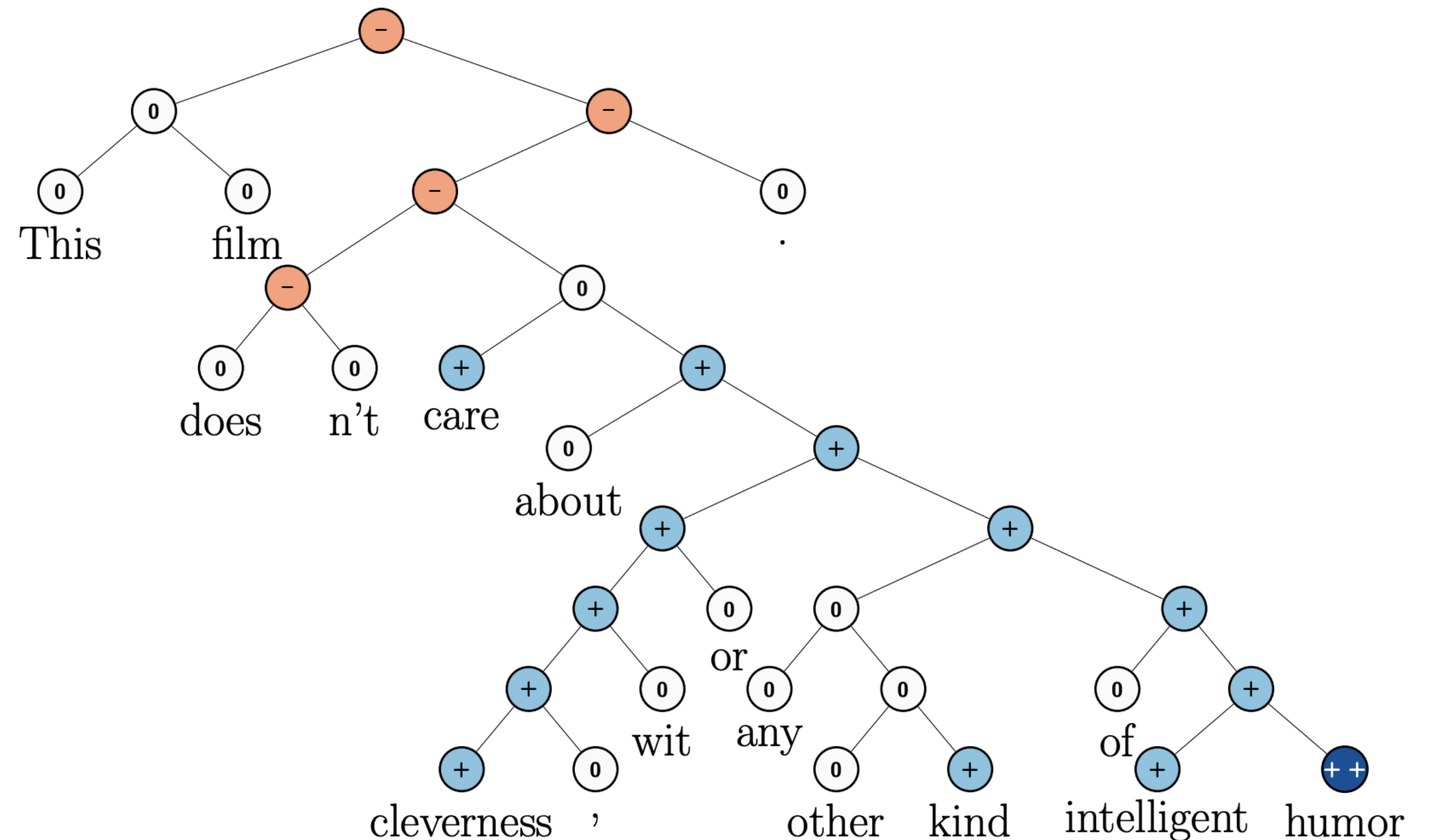
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5 classes

- Very Positive
- Positive
- Neutral
- Negative
- Very Negative

Stanford Sentiment Treebank

This file doesn't care about cleverness, wit or any other kind of intelligent humor.



Recursive deep models for semantic compositionality over a sentiment treebank

Socher et al, EMNLP 2013